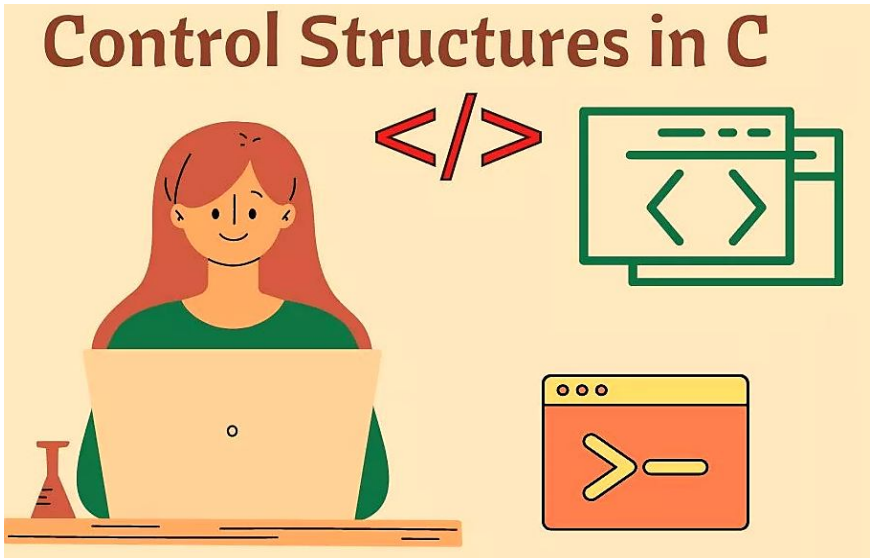
Structured Programming Concepts

(26CSE0105)



Decision Making in C

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Topics to be covered

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- ***Conditional/Decision Making Statements:***
  - ***If statement***
  - ***if-else statement***
  - ***Nested if-else***
  - ***else-if ladder***
  - ***switch-case***

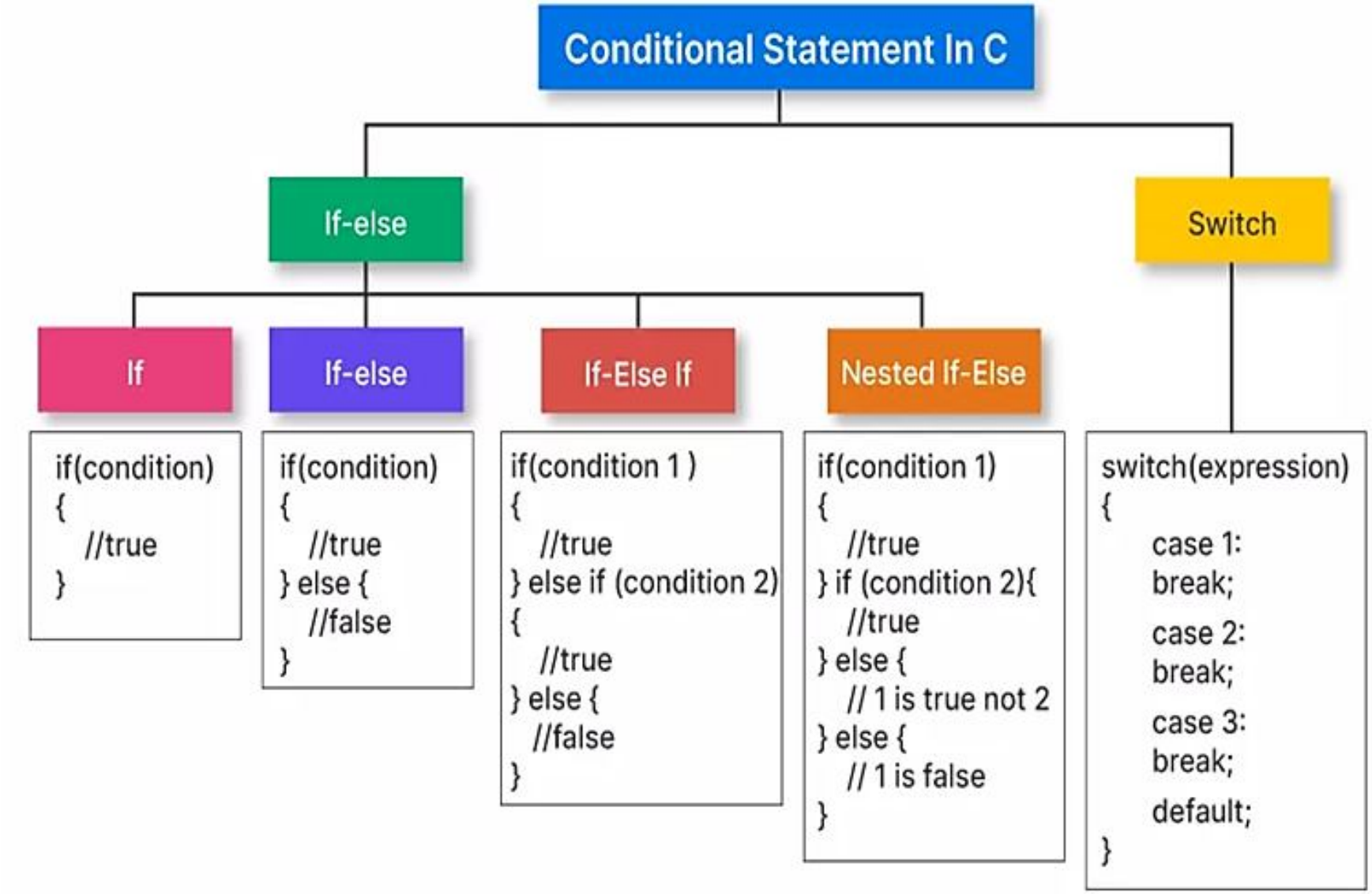
Conditional Statements in C

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- *In practice, we have a number of situations where we may have to **change the order of execution** of statements **based on certain conditions**.*
- *This ability is called **decision making** and the statements used for it are called **conditional statements**.*
- *These statements evaluate **one** or **more conditions** and **make decision** whether to **execute a block of code** or **not**.*

Conditional Statements in C

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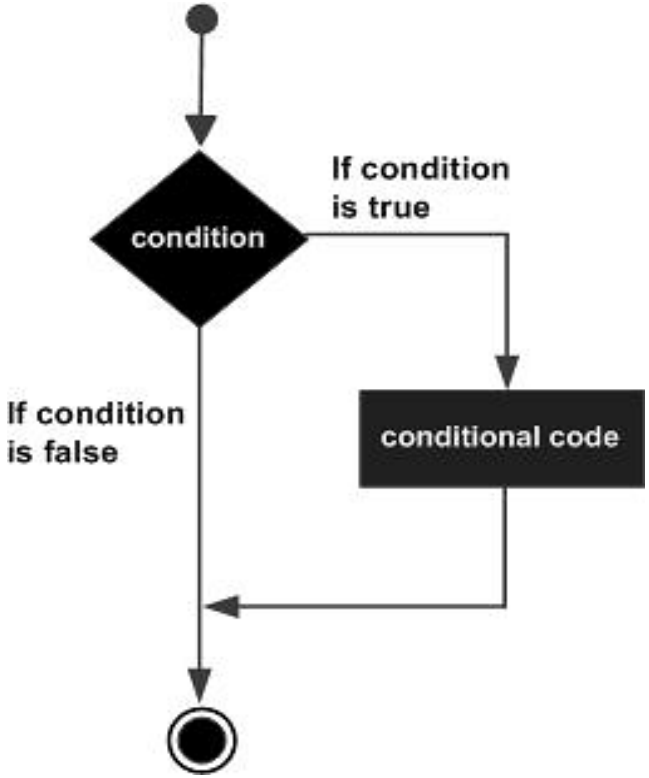


The if statement

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- *It is the **simplest decision-making statement**.*
- *Used to decide whether a certain **statement** or **block of statements** will be **executed** or **not.***
- *If we have **multiple conditions** to check, we can use the **if statement multiple times**.*

```
if (condition)
{
    // block of code to be executed
}
```



if statement (example)

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```
#include <stdio.h>
int main()
{
      int age = 20;
      if (age >= 18)
      {
          printf("Eligible for vote");
      }
}
```

```
Eligible for vote

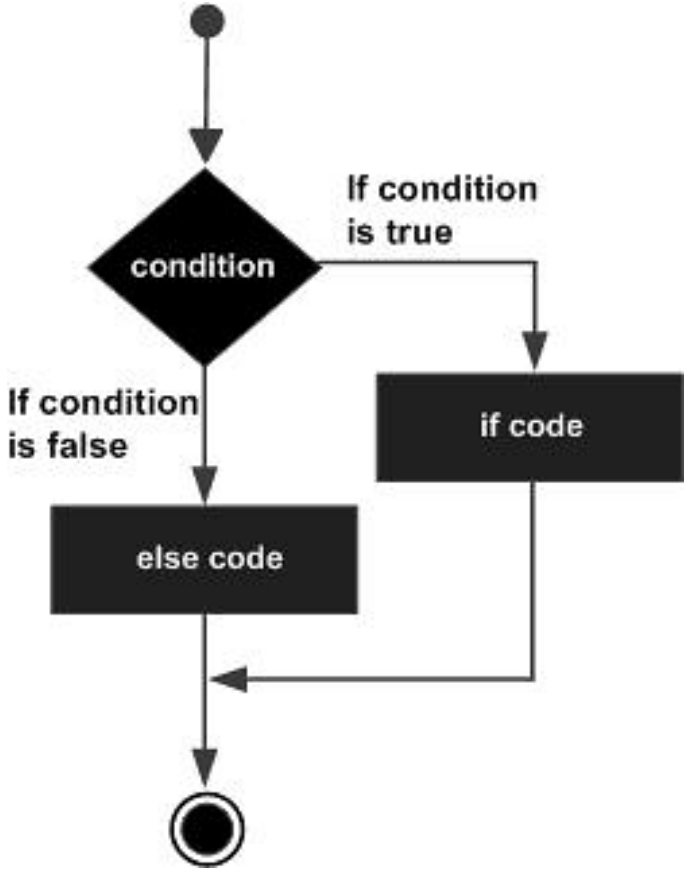
...Program finished with exit code 0
Press ENTER to exit console.
```

The if-else statement

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- *The **if-else statement** offers an **alternative path** when the **condition isn't met**.*

```
if (condition)
{
      // block of code to be executed
}
else
{
     // block of code to be executed
}
```



if-else (example)

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```
#include <stdio.h>
int main()
{
   int age = 10;
   if (age >= 18)
   {
      printf("Eligible for vote");
   }
   else
   {
      printf("Not Eligible for vote");
   }
   return 0;
}
```

```
Not Eligible for vote

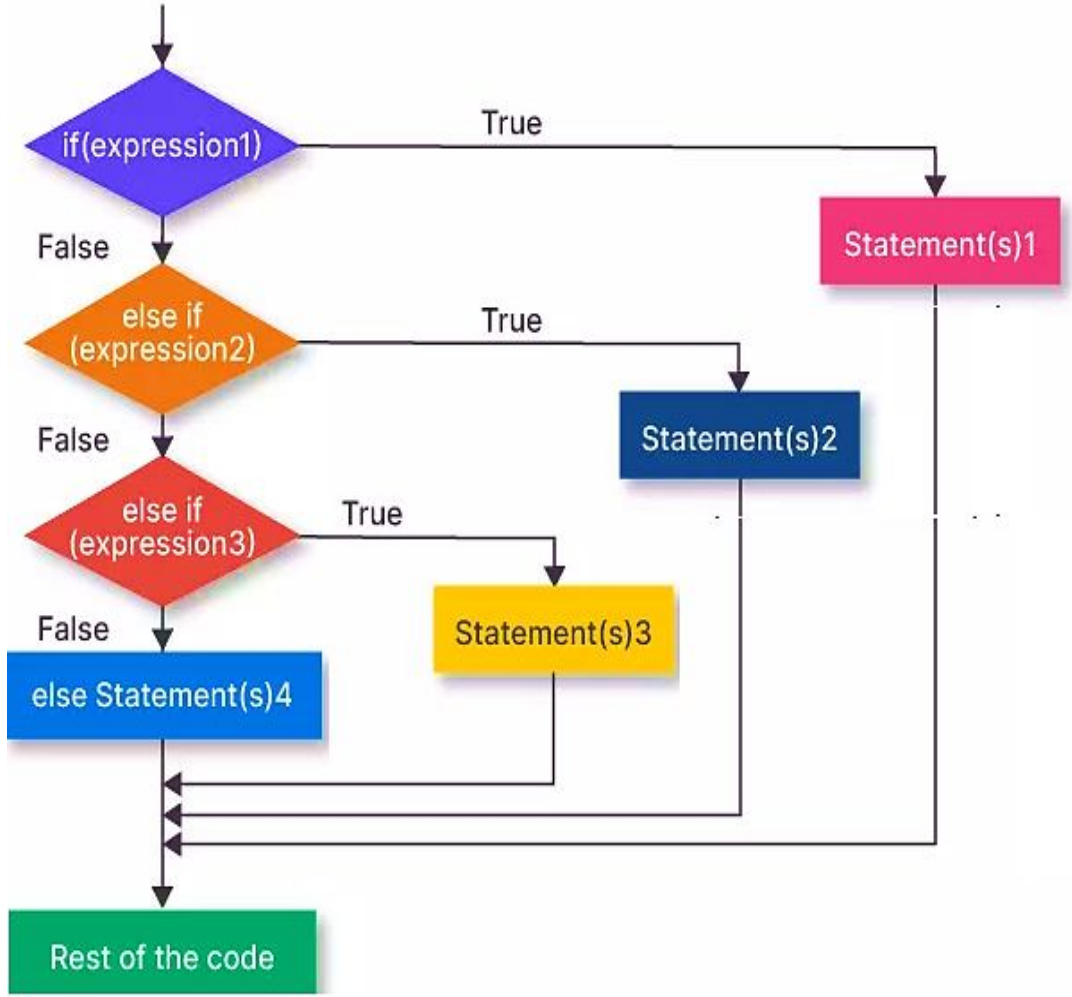
...Program finished with exit code 0
Press ENTER to exit console.
```

The if-else-if statement

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- *The **if-else-if ladder** is an **extension of simple if-else** statement that is used to **test multiple conditions sequentially**.*

```
if (condition1)
{
        // Code to be executed
} else if (condition2)
{
        // Code to be executed
} else if (condition3)
{
        // Code to be executed
} else
{
        // Code to be executed
}
```



if-else-if (example)

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```
#include <stdio.h>
int main()
{
  int items = 72;
    if (items >= 90)
    {
     printf("Store is full");
    }
    else if (items >= 80)
    {
     printf("Need few more items");
    }
    else if (items >= 70)
    {
     printf("Store is partially full");
    }
    else if (items >= 50)
    {
     printf("Store is half full");
    }
    else
    {
     printf("Store is empty");
    }
  return 0;
}
```

```
Store is partially full

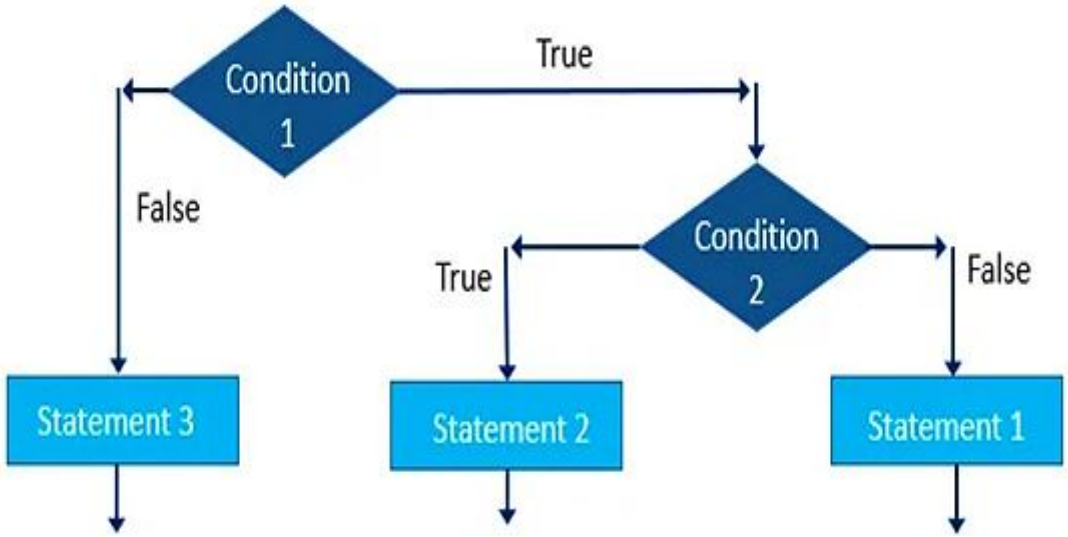
...Program finished with exit code 0
Press ENTER to exit console.
```

The nested if-else statement

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- ***Nested if-else statements*** *mean an* ***if statement*** ***inside another*** ***if statement***.

```
if (condition1)
{
     if (condition2)
    {
            // Code to be executed
    }
    else
    {
            // Code to be executed
     }
else
{
            // Code to be executed
}
```



nested if-else (example)

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```
#include <stdio.h>
int main()
{
   int age = 11;
   if (age >= 18)
   {
      if (age >= 60)
         printf("Eligible to vote (Senior Citizen)\n");
      else
         printf("Eligible for vote\n");
   }
   else
   {
      printf("Not eligible to vote (Under 18)\n");
   }
   return 0;
}
```

```
Not eligible to vote (Under 18)

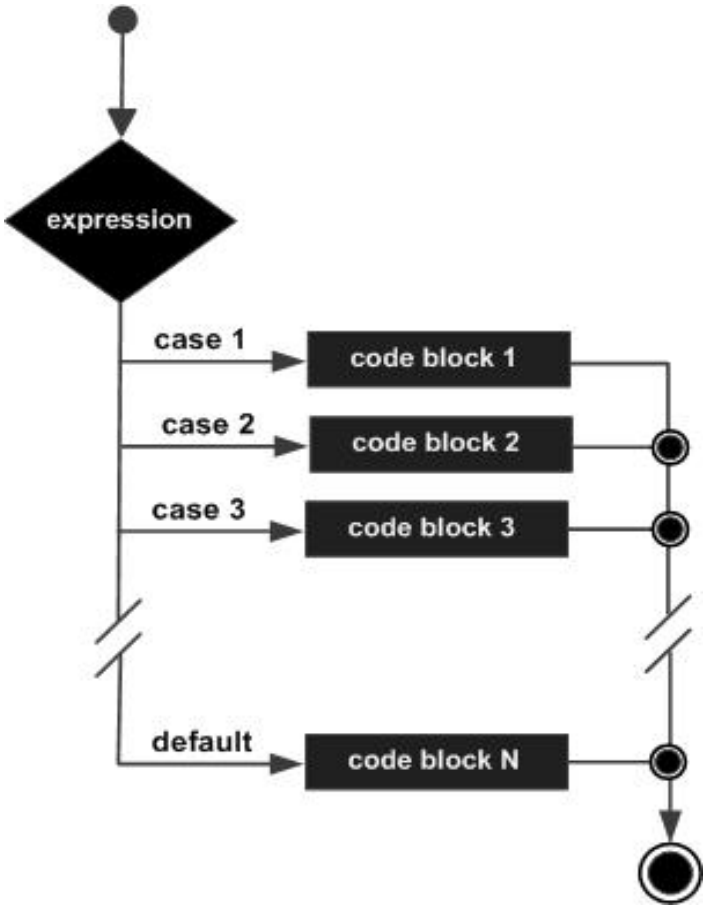

...Program finished with exit code 0
Press ENTER to exit console.
```

The switch-case statement

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- *The **switch case statement** is an **alternative to if else if ladder** or **multi-way decision statement**.*
- *Used to execute the **conditional code** based on the **value** specified in the **switch statement**. **Each value** is called a **case**.*

```
switch (Expression)
{
case Value1:
        Statement1;
        Statement2;
        break;
case Value2:
        Statement1;
        Statement2;
        break;
. .
default:
        Statement1;
        Statement2;
}
```



The switch-case statement

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```
#include <stdio.h>
int main()
{
  char ch;
  printf("Enter time code: ");
  scanf("%c",&ch);
  switch (ch)
  {
    case 'a':
      printf("Good Afternoon\n");
      break;
    case 'e':
      printf("Good Evening\n");
      break;
    case 'm':
      printf("Good Morning\n");
    default:
      printf("Default Case\n");
      break;
  }
  return 0;
}
```

```
Enter time code: e
Good Evening


...Program finished with exit code 0
Press ENTER to exit console.
```

Exercise

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1. *If the ages of Ram, Sham and Ajay are input through the keyboard, write a program to determine the youngest of the three.*

2. *An electric power distribution company charges its domestic consumers as follows:*

Consumption	***Units Rate of Charge***
0 – 200	*Rs. 0.50 per unit*
201 – 400	*Rs. 100 plus Rs. 0.65 per unit excess of 200*
401 – 600	*Rs. 230 plus Rs. 0.80 per unit excess of 400*
601 and above	*Rs. 390 plus Rs. 1.00 per unit excess of 600*

Write a program to enter the units consumed and print the total bill.

Exercise

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3. Write a program to calculate the salary as per the following table:

Gender	***Years of Service***	***Qualifications***	***Salary***
Male	*>= 10*	*Post-Graduate*	*15000*
	>= 10	*Graduate*	*10000*
	< 10	*Post-Graduate*	*10000*
	< 10	*Graduate*	*7000*
Female	*>= 10*	*Post-Graduate*	*16000*
	>= 10	*Graduate*	*11000*
	< 10	*Post-Graduate*	*10000*
	< 10	*Graduate*	*6000*

Enter Gender, Years of Service and Qualifications.
(for PG enter 0 and for G enter 1).

Exercise

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4. A company insures its drivers in the following cases:

- *If the driver is **married**.*
- *If the driver is **unmarried, male** & above **30 years** of age.*
- *If the driver is **unmarried, female** & above **25 years** of age.*
- *In all other cases, the driver is **not insured**.*

If the marital status, sex and age of the driver are the inputs, write a program to determine whether the driver should be insured or not.

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